CSE226:ANDROID APP DEPLOYMENT

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: build responsive Android applications using advanced Jetpack Compose components and modern UI design practices

CO2 :: apply asynchronous and reactive programming techniques using Kotlin Coroutines, Flow, StateFlow and SharedFlow

CO3 :: analyze Android architecture components and implement MVVM based application structures using lifecycle components and state management techniques

CO4 :: build data driven Android applications using Room Database and offline data persistence techniques

CO5 :: build location based Android applications using location services and Google Maps Compose

CO6 :: apply version control, debugging and basic testing techniques to improve Android application quality and maintainability

Unit I

Advanced Jetpack Compose UI Development : Material 3 Components, Bottom Sheets, Dialogs and Alert Dialogs, LaunchedEffect, DisposableEffect, SideEffect, Compose Animation APIs, Custom Composable Design Patterns, UI Performance Optimization, Responsive UI Development

Unit II

Kotlin Coroutines and Reactive Programming : Coroutine Basics, Coroutine Scopes and Context, Dispatchers, Structured Concurrency, Suspending Functions, Cancellation and Timeouts, Exception Handling, Flow, StateFlow, SharedFlow, Flow Operators, Coroutines with Compose

Unit III

Android Architecture Components : MVVM Architecture, View Model, Lifecycle-Aware Components, State Management, State Hoisting, SavedStateHandle, Repository Pattern, Single Source of Truth, UI State and Business State Management

Unit IV

Local Data Persistence and Offline Storage : Room Database Architecture, Entities, Data Access Objects, Relationships, Type Converters, Database Migration, Room with Flow, Data Synchronization Concepts, Offline-First Application Development

Unit V

Location-Based Android Applications : Runtime Permissions, Fused Location Provider, Location Updates, Geocoding and Reverse Geocoding, Google Maps Compose, Markers and Map Customization, Location-Aware Applications, Integration of Location Services with Local Data

Unit VI

Version Control, Debugging and Application Quality : Introduction to Android Application Quality, Basic Unit Testing using JUnit, Basic Jetpack Compose UI Testing, Debugging using Logcat and Android Studio Debugger, Exception and Crash Analysis, Introduction to Version Control, Git Fundamentals, Local Repository Management, GitHub Repository Management, Branching and Merging, Pull Requests, Conflict Resolution, Collaborative Development Workflows

List of Practicals / Experiments:

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- Develop an Android application using Material 3 components and Compose animations.
- Implement side effects and state management in a Compose-based application.
- Create an application using Kotlin Coroutines for asynchronous task execution.
- Develop an application using Flow, StateFlow, and SharedFlow for reactive data handling.
- Design a multi-screen Android application using MVVM architecture and ViewModel.
- Implement Repository Pattern for managing application data.
- Create a Room Database application with CRUD operations.

- Develop an offline-first Android application using Room and Flow.
- Build an application to access and display the user's current location.
- Integrate Google Maps Compose and implement map customization features.
- Write basic unit tests for Android application components using JUnit.
- Perform Compose UI testing for Android screens.
- Debug Android applications using Logcat and Android Studio tools.
- Create and manage a Git repository for an Android project.
- Implement branching, merging, and conflict resolution using Git.

Text Books:

1. BUILDING ANDROID PROJECTS WITH KOTLIN by PANKAJ KUMAR, BPB Publication, New Delhi

References:

1. CLEAN ARCHITECTURE FOR ANDROID - 2ND EDITION by ERAN BOUDJNAH, BPB PUBLICATIONS

2. BUILDING ANDROID PROJECTS WITH KOTLIN by PANKAJ KUMAR, BPB PUBLICATIONS